MIL-P-13044E

19 May 1969
SUPERSEDING
MIL-P-13044D
30 October 1964
(see 6.3)

MILITARY SPECIFICATION

PADDLES, FOOD STIRRING

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope.- This specification covers two sizes of aluminum food-stirring paddles.

* 1.2 Classification.- Paddles shall be of the following sizes as specified (see 6.2):

Size 1 - 42 inches in length.
Size 2 - 55 inches in length.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

QQ-A-225/8	-	Aluminum Alloy Bar, Rod, Wire, and Special Shapes; Rolled, Drawn or Cold Finished, 6061.
QQ-A-250/8	-	Aluminum Alloy 5052, Plate and Sheet.
QQ-A-367	-	Aluminum Alloy Forgings.
QQ-R-566	-	Rods, Welding, Aluminum and Aluminum Alloys.
PPP-B-636	-	Box, Fiberboard.
PPP-F-320	-	Fiberboard; Corrugated and Solid, Sheet Stock (Container Grade), and Cut Shapes.

MILITARY

MIL-A-8625	-	Anodic Coatings, for Aluminum and Aluminum Alloys.

FSC 7330

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STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.
MIL-STD-130 - Identification Marking of U.S. Military Property.
MIL-STD-147 - Palletized anc Containerized Unit Loads 40" X 48" Pallets, Skids, Runners, or Pallet-Type Base.

DRAWING

ARMY NATICK LABORATORIES

2-9-234 - Paddles, Food Stirring; Size 1 and Size 2.

(Miniature copy of Drawing 2-9-234, identified as figure 1 is attached for information purposes only).

(Copies of specifications, standards, and drawing required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications.- The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

Specification B244-62 - Measuring Thickness of Anodic Coating on Aluminum with Eddy Current Instruments.

(Application for copies should be addressed to the American Society for Testing and Materials (ASTM), 1916 Race Street, Philadelphia, Pa. 19103.)

NATIONAL CLASSIFICATION BOARD

National Motor Freight Classification

(Application for copies shall be addressed to the National Classification Board, 1616 P Street, N.W. Washington, D.C. 20036.)

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UNIFORM CLASSIFICATION COMMITTEE

Uniform Freight Classification

(Application for copies shall be addressed to the Uniform Classification Committee, 202 Union Station, Chicago, Ill 60606.)

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

3. REQUIREMENTS

* 3.1 First article.- Unless otherwise specified (see 6.2), before production is commenced, a sample food-stirring paddle shall be submitted or made available to the contracting officer or his authorized representative for approval in accordance with 4.2. Approval of the preproduction sample authorizes commencement of production, but does not relieve the supplier of responsibility for compliance with the detailed provisions of this specification. The preproduction sample shall be manufactured in the same facilities to be used for the manufacture of the production items.

3.2 Materials.- Unless otherwise specified herein, commercial tolerances are applicable to all metal thicknesses and diameters. Materials not definitely specified shall be of the quality normally used by the manufacturer for paddles food stirring provided the completed item complies with all provisions of the specification.

* 3.3 Design and construction.- The design and construction of the food-stirring paddles shall conform to the details and dimensions shown on Drawing 2-9-234 and as specified herein. The finished paddle shall have no warping or twisting exceeding 1/8 inch. The paddles shall be capable of being dropped 50 times on the tip and from a height of 30 inches without deformation or distortion exceeding 1/8 inch when tested as specified in 4.4.1. The handles shall be welded at the cap end and blade end to provide water tight joints (see 4.4.2).

* 3.3.1 Blades.- The blades shall be welded or forged construction and shall conform to shape and dimensions shown on Drawing 2-9-234.

* 3.3.1.1 Welded blades.- The welded blades shall be fabricated from aluminum alloy sheet conforming to QQ-A-250/8. The beaded edge shall be fabricated from aluminum alloy rod conforming to QQ-A-225/8.

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3.3.1.2 Forged blade.- The forged blades shall be 6061 aluminum alloy conforming to QQ-A-367. The weld joining the forged blade to the handle shall be made using type I, class 4043 rod conforming to QQ-R-566.

3.3.1.3 Wood insert.- The wood insert shall be maple or any equivalent hardwood. The wood insert shall be inclosed in the handle of the paddles as shown on Drawing 2-9-234. The wood insert shall be one-piece extending the full length of the handle, and into the extension of the handle in the blade.

3.4 Welding.- Surface of parts to be welded shall be cleaned of scale, grease, or other foreign matter. Welds shall be sound, smooth, with no pits and flashes. Welds shall not be porous, cracked, or not fused. Welded joints shall be ground. There shall be no evidence of weld failure when tested as specified in 4.4.1.

3.5 Finish.- The surfaces of the paddles shall be cleaned of scale, grease, or other foreign matter. There shall be no stain or foreign matter imbedded in the finish. The paddles shall have an anodized finish in accordance with type II process specified in MIL-A-8625. Anodizing shall be a minimum thickness of 0.850 mils with tested as specified in 4.4.3. The anodized surfaces shall be buffed or etched. The finish shall not have scratches, crevices, or deep holes.

3.6 Marking for identification.- The paddles shall have the manufacturer's name, tradename, or trademark of such known characters as to be easily identifiable with the manufacturer. Marking shall be applied in accordance with MIL-STD-130.

3.7 Workmanship.- The finished paddles shall be clean and without fractures, splits, or malformation.

4. QUALITY ASSURANCE PROVISIONS

* 4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless otherwise disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

* 4.2 First article inspection.- When a preproduction sample is required, it shall be examined for defects listed in table I and dimensions specified, and tested as specified in 4.4.1, 4.4.2, and 4.4.3. Failure to pass the test in 4.4.1, 4.4.2 and 4.4.3 shall be cause for rejection of the preproduction sample. The presence of any defects listed in table I or any dimension not within specified requirements shall be cause for rejection of the preproduction sample.

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* 4.3 Inspection.- Sampling for inspection shall be performed in accordance with MIL-STD-105, except where otherwise indicated hereinafter.

* 4.3.1 Component and material inspection.- In accordance with 4.1 above, components and materials shall be inspected and tested in accordance with all the requirements of referenced specifications, drawings, and standards unless otherwise excluded, amended, modified or qualified in this specification or applicable purchase document.

4.3.2 Intermediate dimensional examination.- Examination shall be made of the wood insert, handle, and blade as applicable, for compliance with dimensions shown on Drawing 2-9-234, which cannot be checked on the end item. The sample unit shall be one wood insert, one handle, or one blade as applicable. The inspection lot shall be the number of wood inserts, handles, or blades offered for inspection at one time. The inspection level shall be S-3 with an acceptable quality level (AQL) of 2.5 defects, expressed in terms of defects per hundred units.

4.3.3 In-process examination.- Examination shall be made to determine that surfaces of welded joints conform to 3.4 and that anodizing conforms to 3.5. Whenever nonconformance is noted, corrections shall be made on the affected items and process.

4.3.4 End product inspection.- The inspection lot shall consist of all paddles of one size offered for inspection at one time. The sample unit for this inspection shall be one completely fabricated and assembled paddle.

4.3.4.1 Visual examination.- Examination of the end item shall be in accordance with the classification of defects in table I. The inspection level shall be level II with an AQL of 2.5 for major defects and 6.5 for total defects, expressed in terms of defects per hundred units.

TABLE I.- Classification of defects

Examine	Defect	Classification Major	Minor
Finish	Sharp edge, corner, burr, or sliver	X	
	Stained or foreign matter imbedded in finish	X	
	Not buffed or etched	X	
	Pits, scratches, crevices, or holes deep enough to hold food particles or dirt	X	

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TABLE I.- Classification of defects (cont'd)

Examine	Defect	Classification Major	Minor
Construction, design, and workmanship	Any component missing	X	
	Any component fractured, split, or malformed	X	
Welding	Welds contain pits or flashes	X	
	Welded joints not ground	X	
	Missing, porous, cracked, or not fused	X	
Identification marking	Missing, incomplete, not legible		X

4.3.4.2 Dimensional examination.- Examination shall be made of the end item to determine compliance with this specification. Any dimension that is not within the specified requirements shall be classified a defect. The inspection level shall be S-2 with an AQL of 4.0 defects, expressed in terms of defects per hundred units.

4.3.4.3 Testing of the end item.- The end item shall be tested as specified in 4.4.1, 4.4.2, and 4.4.3, using an inspection level of S-2 with an AQL of 4.0 defects, expressed in terms of defects per hundred units.

4.3.5 Examination of preparation for delivery.- An examination shall be made to determine that packaging, packing, and marking as required by section 5 of this specification are complied with. Defects shall be scored as specified in table II. The sample unit shall be one shipping container fully prepared for delivery. The lot shall be the number of containers offered for inspection at one time. The inspection level shall be S-2 with an AQL of 4.0 defects, expressed in terms of defects per hundred units.

TABLE II.- Preparation for delivery defects

Examine	Defect
Markings, exterior and interior	Omitted; incorrect; illegible; of improper size, location, sequence, or method of application.
Materials	Component missing, damaged, or otherwise defective.
Content	Number per container is more or less than required.

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4.3.5.1 Examination of shipping containers.- When shipping containers are required to be in accordance with PPP-B-636, examination for defects in closure, waterproofing, and strapping shall be in accordance with PPP-B-636.

4.4 Tests.-

4.4.1 Drop test.- The paddles shall be dropped vertically 50 times on the tip end of the blade from a height of 30 inches onto a steel plate to determine compliance with deformation and distortion requirements of 3.3. The welds shall be checked to determine compliance with 3.4. Failure to comply with 3.3 and 3.4 shall constitute failure of this test.

4.4.2 Water immersion test.- The paddles shall be completely immersed in water for a period of 5 minutes. The temperature of the water shall be 125 degrees Fahrenheit (F.), plus or minus 5 degrees F. The welded joints of the handle shall be observed during this period for emission of air bubbles to determine compliance with the water tightness requirements of 3.3. Failure to comply with 3.3 shall constitute failure of this test.

4.4.3 Thickness test for anodic coating.- The anodic coating thickness of the paddle shall be checked in accordance with ASTM B244-62 to determine compliance with 3.5. Failure to comply with 3.5 shall constitute failure of this test.

5. PREPARATION FOR DELIVERY

5.1 Packaging.- Packaging shall be level A or C as specified (see 6.2).

5.1.1 Level A.- Two paddles of one size only, shall be packaged with ends alternated in a snug-fitting fiberboard box conforming to style FPF, type CF, variety SW, or type SF, class domestic, grade 200 of PPP-B-636. The paddles shall be separated by a fiberboard pad made of the same material as the box, cut one-eigth of an inch less than the inside length and width of the box. Box closure shall be in accordance with the appendix of PPP-B-636.

5.1.2 Level C.- The paddles shall be packaged to afford adequate protection against damage during shipment for the supply source to the first receiving activity. The supplier may use his standard practice when it meets this requirement.

5.2 Packing.- Packing shall be level A, B, or C as specified (see 6.2).

5.2.1 Level A.- Twenty size 1 or ten size 2 paddles, packaged as specified in 5.1, shall be packed in a snug-fitting fiberboard shipping container conforming to style RSC, V2s of PPP-B-636. Each shipping container shall be closed, waterproofed by means of tape, and reinforced with flat strapping or tape banding in accordance with the appendix of PPP-B-636.

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5.2.2 Level B.- Twenty size 1 or ten size 2 paddles, packaged as specified in 5.1, shall be packed in a snug-fitting fiberboard shipping container conforming to style RSC, type CF, variety SW, or type SF, class domestic of PPP-B-636. Each shipping container shall be closed in accordance with method II of the appendix of PPP-B-636.

5.2.2.1 When specified (see 6.2), the shipping container shall be V3c or V3s, fabricated and closed in accordance with PPP-B-636. The shipping container material may also be grade V4s of PPP-F-320.

5.2.3 Level C.- Paddles, packaged as specified in 5.1, shall be packed in a manner to insure carrier acceptance and safe delivery at destination at the lowest transportation rate for such supplies. Containers shall be in accordance with Uniform Freight Classification Rules or National Motor Freight Rules, as applicable.

5.3 Palletization.- Unless otherwise specified (see 6.2), paddles packed as specified shall be palletized in accordance with load type I of MIL-STD-147. Each prepared load shall be bonded with primary and secondary straps in accordance with bonding means K and L. Pallet patterns shall be in accordance with the appendix of MIL-STD-147. Interlocking of loads shall be effected by reversing the pattern of each course. If the container is of a size which does not conform to any of the pallet patterns specified in MIL-STD-147, the pallet pattern used shall be approved by the contracting officer.

5.4 Marking.- In addition to any special marking required by the contract or order, interior packages and shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use.- The paddles covered by this specification are intended for use in Military kitchens for stirring food. The size 1 paddle is designed for use with 20, 30, and 40 gallon capacity kettles. The size 2 paddle is designed for use with 60, 80, and 100 gallon capacity kettles.

* 6.2 Ordering data.- Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Size required (see 1.2).
(c) First article.- When preproduction sample is waived (see 3.1).
(d) Selection of the applicable levels of packaging and packing (see 5.1 and 5.2).
(e) When weather-resistant grade fiberboard shipping containers are required for level B packing (see 5.2.2.1).
(f) When palletization is not required (see 5.3).

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6.3 Supersession data.- The style 1 paddle covered by MIL-P-13044D, dated 30 October 1964, has been deleted herein. Size 1 covered herein is the same length as the former style 1 and size 2 is the same as the former style 2.

6.4 The margins of this specification are marked with an asterisk (*) to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and suppliers are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Custodians:

Army - GL
Navy - SA
Air Force - 84

Preparing activity:

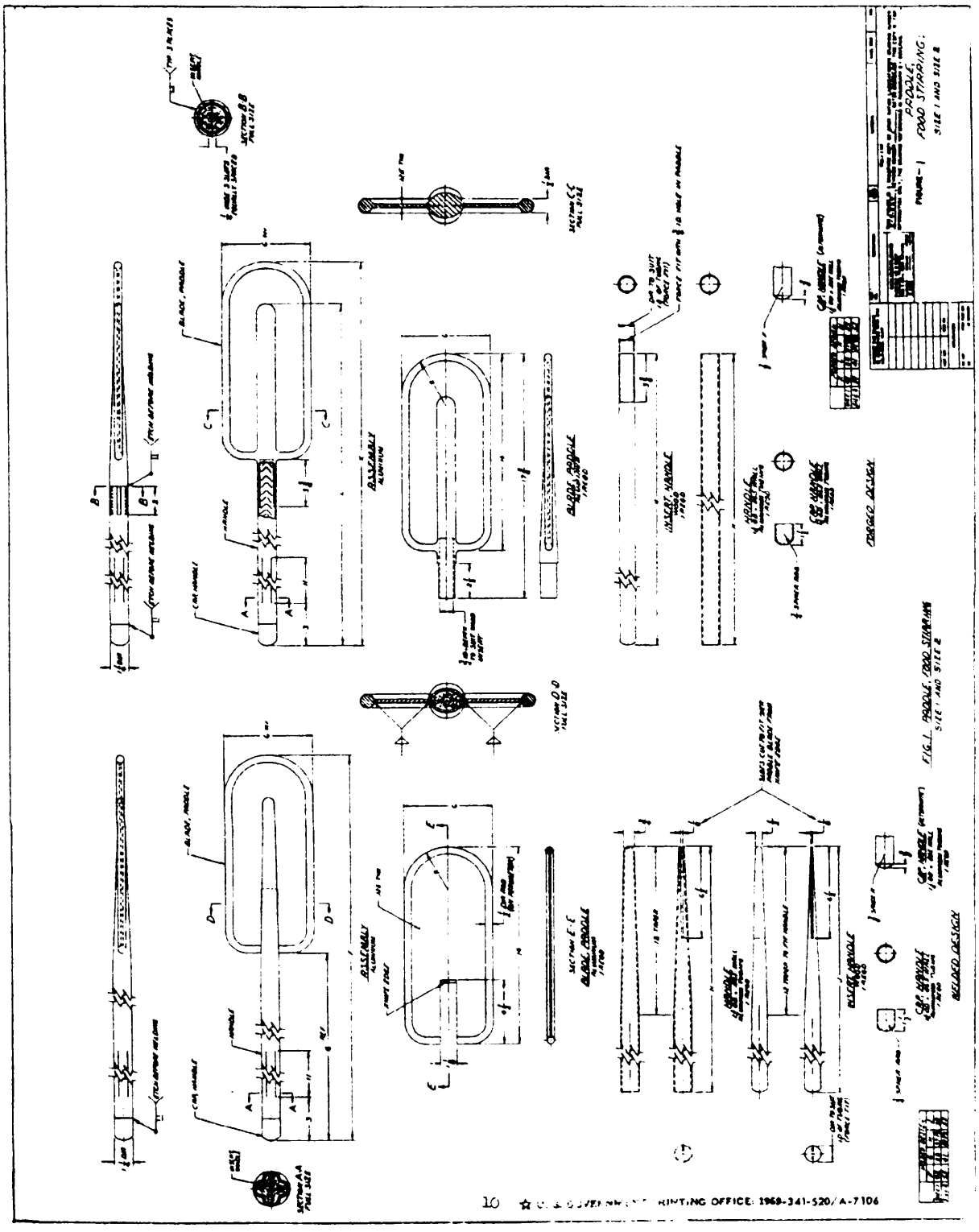
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Review activities:

Army - MD
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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

MIL-P-13044E Paddles, Food Stirring

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD FORM 1 JAN 66 1426

REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

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